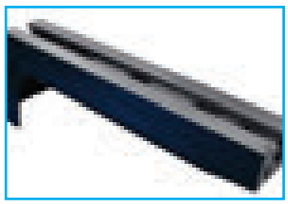
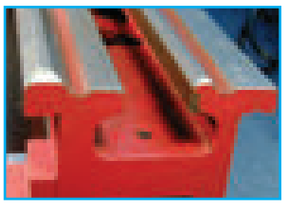
The top of the bed has a guideways which is manufactured in two shapes

1. Flat shaped
2. V – Shaped



Flat shaped

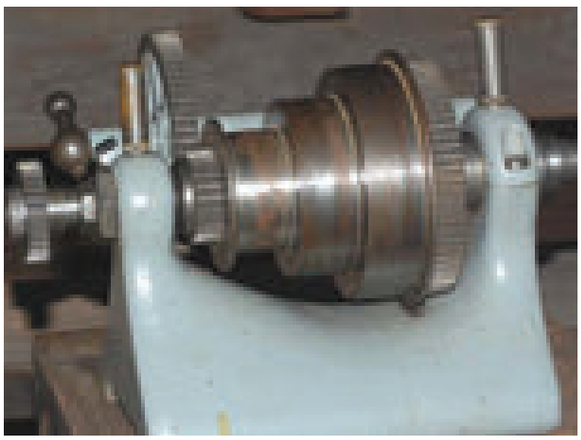


V – Shaped

The tailstock and the carriage slides on these guideways. Flat bed guide ways can be found in older machine tools. It is useful in heavy duty machines, handling large workpieces. Though, it is less accuracy. V-type bed have been found in modern machine tools. This type of V-bed gives accurate alignment to carriage and tailstock. The metal chips automatically fall through.

## Headstock

Headstock is mounted permanently on the inner guideways at the left hand side of the lathe bed. The headstock houses a hollow spindle and the mechanism for driving the spindle at multiple speeds. The headstock will have any of the following arrangements for driving and altering the spindle speeds.

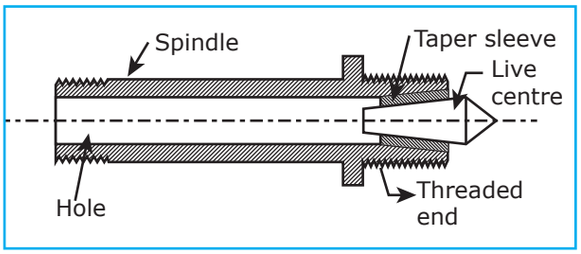


Headstock

1. Stepped cone pulley mechanism
2. Back gear mechanism
3. Gear box mechanism

## Spindle

The spindle freely rotates on two large bearings housed on the headstock casting. It is made of carbon or nickel chrome steel. A hole extends through the spindle so that a long bar stock may be passed through the hold. The front end of the spindle is threaded on which any one of the work holding devices can be screwed. The front end of the hole is tapered to receive live centre which supports the work. On the other side of the spindle, a gear known as a spindle gear is fitted. Through this gear, tumbler gears and a main gear train, the power is transmitted to the gear on the lead screw.




Spindle

## Tailstock

Tailstock is located on the inner guideways at the right side of the lathe bed. The body of the tailstock is bored and houses the tailstock spindle. The spindle moves front and back inside the hole. The spindle has a tapper hole to receive the dead centre or shanks of tool like drill or reamer. If the tailstock hand wheel is rotated in clockwise direction, the spindle advances. The spindle will be reversed inside the hold, if the handwheel is rotated in anti-clockwise direction.



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